

A distributional analysis of QuickXsort for Mergesort

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Florian Lesny and Ralph Neininger (both Frankfurt)

June 25, 2026

QuickMergeSort

- Sorting: Given a list of n elements $x_1, \dots, x_n \in \mathbb{R}$, return the sorted list
- One of the most basic and most used algorithms
- MergeSort is a fast sorting algorithm (only $n \log_2 n + O(n)$ comparisons), but it needs $\Theta(n)$ additional storage
- QuickSort does not need much additional storage, but needs $n \ln n + O(n)$ comparisons on average
- Idea: Can we combine MergeSort with QuickSort to not need so much additional storage?
- This is QuickMergeSort
- Aim: Show limiting distribution of the number of comparisons of QuickMergeSort in the uniform model and study this limit

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- Limit distribution is an open question posed by Wild last AofA
- Edelkamp, Weiß, and Wild (2020) derived, among other things, the asymptotic mean

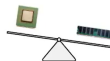
Multiway Quicksort Summary

Multiway Quicksort

- 1 tricky **asymmetries** in YBB Quicksort


 saves a few comparisons

- 2 evidence for growing weight of memory transfer costs



- 3 **Multiway Quicksort** reduces **memory transfer**



- No other optimization of Quicksort achieves that

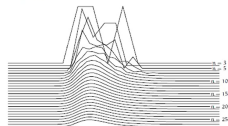
- 4 AofA for Algorithm Science!

Open Problem

- Automated/Convenient way to determine linear term of cost?

Much more to tell ...

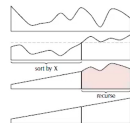
- Limiting distributions of costs



- **Sesquiselect** (cache-efficient selection)



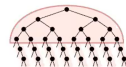
- **QuickXsort** inplace unstable mergesort



Open Problem

- limit distribution?

- Intriguing interplay of parameters



QuickXSort

- Generally, **QuickXSort** is a method that combines a sorting algorithm **XSort** with QuickSort:

$$\begin{aligned} \text{SortingAlgorithms} &\rightarrow \text{SortingAlgorithms}, \\ \text{XSort} &\mapsto \text{QuickXSort} \end{aligned}$$

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QuickXSort

- Randomly (with some assumptions on distribution*) select an element as **pivot**
- Partition in elements smaller and larger than the pivot
- Use **XSort** on the bigger list if possible** with the other as buffer, permutating it
- Recurse: Use **QuickXSort** on the smaller list



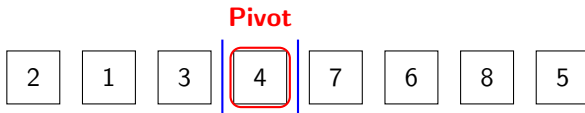
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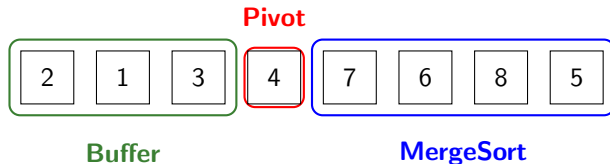
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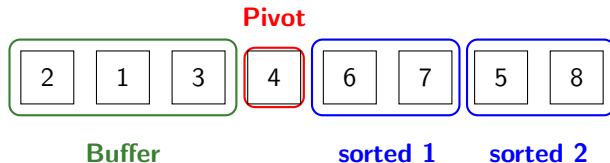
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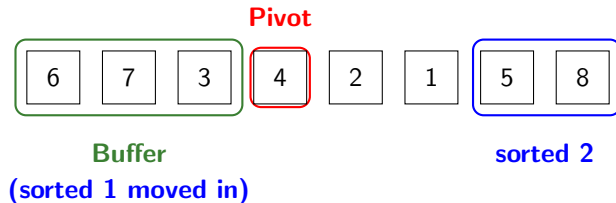
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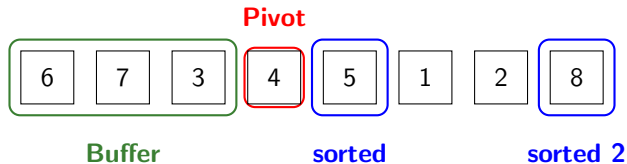
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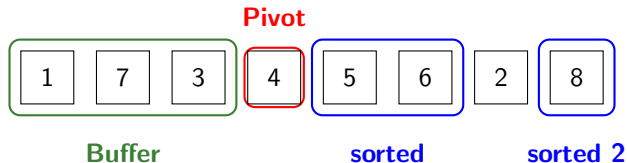
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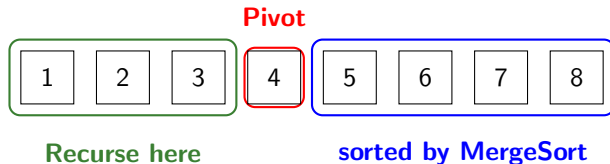
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Assumptions on Pivots and XSort

Pivot Selection

Assume:

- elements $x_1, \dots, x_n$ are distinct,
- the order of elements is uniform in S_n ,
- pivot selection takes $o(n)$ time,
- relative pivot rank converges:

$$\frac{\#\{\text{elements smaller than pivot}\}}{n} \xrightarrow{d} Z'. \quad (\text{P}')$$

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Examples

- **random pivot**, first element or last element:

$$Z' \sim \text{Unif}(0, 1),$$

- **median-of- $(2h - 1)$** : Sample $2h - 1$ elements and take the median

$$Z' \sim \text{Beta}(h, h).$$

- $\text{Beta}(h, h)$ is more concentrated around $\frac{1}{2}$ for higher h

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XSort

- We assume that XSort needs a buffer of size $\sim \alpha n$ for $\alpha \leq 1$ and that it only permutes this buffer
- In order to do asymptotic analysis on QuickXSort, we need asymptotic analysis on XSort.
- Let M_n be the number of comparisons XSort needs to sort n elements in a uniformly random order.

- We assume

$$\begin{aligned}\mathbb{E}[M_n] &= n \log_2 n + \Pi(\log_2 n)n + o(n) \\ \text{Var}(M_n) &= O(n).\end{aligned}\tag{X}$$

for some 1-periodic, continuous function Π .

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- (X) is true for Top-Down MergeSort, Hwang (1998), with

$$\Pi^{td}(x) = \frac{1}{2} - \frac{1}{\log(2)} + \frac{\Psi'(0)}{\log(2)} + \frac{1}{\log(2)} \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{1 + \Psi(\chi_k)}{\chi_k (\chi_k + 1)} e^{2k\pi i x}$$

with $\chi_k = \frac{2k\pi i}{\log(2)}$ and

$$\Psi(u) = \sum_{m \geq 1} \frac{2}{(m+1)(m+2)} \left(-\frac{1}{(2m)^u} + \frac{1}{(2m+1)^u} \right)$$

- and for QueueMergeSort (W.-M. Chen, Hwang, and G.-H. Chen 1999) with

$$\Pi^q(x) = 1 - \{x - \beta\} - 2^{1-\{x-\beta\}}$$

for $\beta = \sum_{j>0} \frac{1}{2^{j+1}}$ and $\{x\} := x - \lfloor x \rfloor$.

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Other XSort

- For the third common MergeSort, Bottom-Up MergeSort, (X), specifically $\text{Var}(M_n) = O(n)$ is not true
 - For $n = 2^m + 1$, the last element is inserted into a sorted 2^m long list, taking $\approx \text{unif}\{1, \dots, n\}$ time.
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Main result and Corollaries

Theorem (I., Lesny, Neininger (2026))

Assume $(\mathbf{P}')$, $(\mathbf{X})$. For the number X_n of key comparisons required by QuickXSort, we have with Υ being a random 1-periodic continuous process that

$$\ell_2 \left(\frac{X_n - n \log_2 n}{n}, \Upsilon(\{\log_2 n\}) \right) \rightarrow 0 \quad (n \rightarrow \infty). \quad (1)$$

ℓ_2 convergence implies convergence in distribution and that, as $n \rightarrow \infty$,

$$\begin{aligned} \mathbb{E}[X_n] &= n \log_2(n) + \Lambda_E(\log_2 n)n + o(n), \\ \text{Var}(X_n) &= \Lambda_V(\log_2 n)n^2 + o(n^2), \end{aligned}$$

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The buffer size

- How big is the buffer?
- For Mergesort, two lists must be merged in every step.
- With the normal implementation, the smaller of these is moved into the buffer, thus requiring $\lfloor \frac{n}{2} \rfloor$ of buffer space
- This can be improved to $\frac{n}{3}$ (Ping-Pong merge) and even further
- Remember that we assume that **XSort** needs a buffer of size $\sim \alpha n$ for some $\alpha \leq 1$.
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 - If we split $\frac{1}{4}$ to $\frac{3}{4}$, then MergeSort can only run on the $\frac{1}{4}$.
- Then, together with (P'),

$$\frac{\text{\# elements in next step of QuickXSort}}{n} \rightarrow Z \quad (P)$$

in distribution

- This Z is

$$Z = \begin{cases} Z' & \text{if } \frac{\alpha}{1+\alpha} < Z' \leq \frac{1}{2} \text{ or } \frac{1}{1+\alpha} < Z' \leq 1 \\ 1 - Z' & \text{else.} \end{cases}$$

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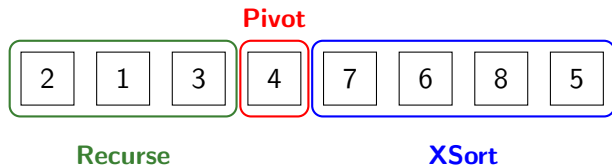
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Contraction Method

The Recursion

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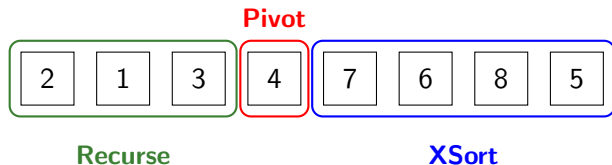
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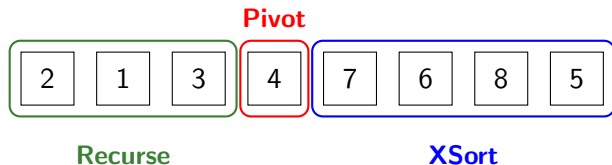
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The Recursion limit

- By (P), $I/n \xrightarrow{d} Z$
- The $n \log_2 n$ terms and n give an entropy-like term

$$\mathcal{E}(Z) := Z \log_2 Z + (1 - Z) \log_2(1 - Z) + 1 \geq 0$$

- This is 1 minus the entropy of a Bernoulli with prob. Z
- $\Rightarrow$ Minimal and 0 at $\frac{1}{2}$, 1 at 0 and 1
- The limit of the recursion is

$$Y_n \stackrel{d}{=} ZY_I + \mathcal{E}(Z) + (1 - Z)\Pi(\log_2((1 - Z)n))$$

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The periodic limit

In

$$Y_n \stackrel{d}{=} ZY_I + \mathcal{E}(Z) + (1 - Z)\Pi(\log_2((1 - Z)n))$$

the limit is still dependant on n , but one can complete it to a full periodic function Υ

$$\Upsilon(x) \stackrel{d}{=} Z\Upsilon + \mathcal{E}(Z) + (1 - Z)\Pi(x + \log_2(1 - Z))$$

Properties of the Limit

The expectation

From

$$\Upsilon(x) \stackrel{d}{=} Z\Upsilon + \mathcal{E}(Z) + (1-Z)\Pi(x + \log_2(1-Z))$$

we see

$$\begin{aligned}\int_0^1 \mathbb{E}\Upsilon(x)dx &= \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \int_0^1 \Pi(x)dx \\ \max(\mathbb{E}\Upsilon) &\leq \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \max(\Pi) \\ \min(\mathbb{E}\Upsilon) &\geq \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \min(\Pi)\end{aligned}$$

So we want Z small, but bigger than $\frac{\alpha}{1+\alpha}$ and close to $\frac{1}{2}$ (so that $\mathcal{E}(Z)$ small), hence we use multiple pivots.

The expectation

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$$\Upsilon(x) \stackrel{d}{=} Z\Upsilon + \mathcal{E}(Z) + (1-Z)\Pi(x + \log_2(1-Z))$$

we see

$$\begin{aligned}\int_0^1 \mathbb{E}\Upsilon(x)dx &= \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \int_0^1 \Pi(x)dx \\ \max(\mathbb{E}\Upsilon) &\leq \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \max(\Pi) \\ \min(\mathbb{E}\Upsilon) &\geq \frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}Z} + \min(\Pi)\end{aligned}$$

So we want Z small, but bigger than $\frac{\alpha}{1+\alpha}$ and close to $\frac{1}{2}$ (so that $\mathcal{E}(Z)$ small), hence we use multiple pivots.

Expectation

Normal Top-down MergeSort has

$$\mathbb{E}[M_n] = n \log_2 n - (1.25265 \pm 0.01185)n + o(n)$$

Corollary (Edelkamp, Weiß, and Wild (2020))

For the X_n from Theorem 1 using Top-down Mergesort, we have

$$\mathbb{E}[X_n] = n \log_2 n - (0.3407 \pm 0.01185)n + o(n) \quad \text{for median-of-1,}$$

$$\mathbb{E}[X_n] = n \log_2 n - (0.8477 \pm 0.01185)n + o(n) \quad \text{for median-of-3.}$$

For Queue-Mergesort,

$$\mathbb{E}[X_n] = n \log_2 n - (0.0450 \pm 0.04303)n + o(n) \quad \text{for median-of-1,}$$

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Variance

- Because the variance for **XSort** is assumed to be $O(n)$, the variance is dominated by the pivot selection
- Additional variance from evaluating Π at a random place $x + \log_2(1 - Z)$
- If Π was constant,

$$\text{Var}(\Upsilon) = \frac{\text{Var}\left(\frac{\mathbb{E}[\mathcal{E}(Z)]}{1 - \mathbb{E}[Z]}Z + \mathcal{E}(Z)\right)}{1 - \mathbb{E}[Z^2]}.$$

- For median-of-3 pivot ($h = 2$) and $\alpha = \frac{1}{2}$, this is roughly 0.1068.

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Fourier coefficients

- Υ is a periodic function, so study its Fourier coefficients
- The k -th Fourier coefficient $c_{\Upsilon,k}$ of Υ fulfils the recursion

$$c_{\Upsilon,k} \stackrel{d}{=} Z^{1+\chi_k} c_{\Upsilon,k} + \mathcal{E}(Z) \mathbf{1}(k=0) + (1-Z)^{1+\chi_k} c_{\Pi,k}.$$

- From this, explicit forms for the Fourier coefficients of mean and variance can be obtained
- The Fourier coefficients (except the 0-th) of the mean are smaller or equal than those of Π , implying the $\mathbb{E}[\Upsilon]$ is smoother than Π .

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Conclusion

- In QuickMergeSort, you pay a small expected $\approx 0.4n$ of speed to not require extra buffer space + $\approx 0.3n$ std deviation
- We identified mean, variance and limit distribution of the number of comparisons in QuickMergeSort
- Mean and variance are both oscillate with $\log_2 n$
- Open Question: If we take as pivot the median of a sample whose size increases in n , $Z \rightarrow \frac{1}{2}$ and $\Upsilon = \Pi$ has no variance. What is then the limit distribution?

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Thanks